Algorithm Sample 5

Dataset-expansion sample

Procedure

The pseudocode below is drawn from a source-backed image-to-LaTeX benchmark and reproduced verbatim.

function TRANSPORT($\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}, \Delta t$) ▷ *Input moments, natural parameters, and gauge parameters for all cells*
  $\boldsymbol{\beta} \leftarrow$ NEWTONOPTIMIZATION($\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{\varphi}(\boldsymbol{u}; \boldsymbol{g})$) ▷ *Solve the natural parameters by Alg. 1 in the gauge* $\boldsymbol{g}$
  $\boldsymbol{F} \leftarrow$ COMPUTEFLUXES($\boldsymbol{\beta}, \boldsymbol{\varphi}(\boldsymbol{u}; \boldsymbol{g})$) ▷ *Compute fluxes of the statistics* $\boldsymbol{\varphi}(\boldsymbol{u}; \boldsymbol{g})$ *by (3)*
  $\boldsymbol{M}_{\pm 1}, \boldsymbol{F}_{\pm 1} \leftarrow$ SPATIALGAUGETRANSFORMATION($\boldsymbol{M}, \boldsymbol{F}, \boldsymbol{g}$) ▷ *Perform gauge transformation as in (4)*
  $\boldsymbol{M} \leftarrow$ FINITEVOLUMESTEP($\boldsymbol{M}, \boldsymbol{F}, \boldsymbol{M}_{\pm 1}, \boldsymbol{F}_{\pm 1}, \boldsymbol{g}, \Delta t$) ▷ *One step forward in time by (5)*
  return $\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}$
end function
function COLLISION($\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}, \Delta t$) ▷ *Input moments, natural parameters, and gauge parameters for all cells*
  $\boldsymbol{M} \leftarrow$ SOURCETERM($\boldsymbol{M}, \boldsymbol{g}, \Delta t$) ▷ *Compute source term by operator splitting (6)*
  return $\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}$
end function
function STEP($\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}$) ▷ *Input moments, natural parameters, and gauge parameters for all cells*
  $\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g} \leftarrow$ COLLISION($\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}, \Delta t/2$) ▷ *Compute collision term by operator splitting*
  $\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g} \leftarrow$ TRANSPORT($\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}, \Delta t$) ▷ *Compute Transport term by operator splitting*
  $\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g} \leftarrow$ COLLISION($\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}, \Delta t/2$) ▷ *Compute collision term by operator splitting*
  $\boldsymbol{g}_H \leftarrow$ COMPUTEGAUGEPARAMETERS($\boldsymbol{M}, \boldsymbol{g}$) ▷ *Compute the Hermite gauge parameters by (7)*
  $\boldsymbol{M}_H, \boldsymbol{\beta}_H \leftarrow$ GAUGETRANSFORMATION($\boldsymbol{M}, \boldsymbol{\beta}, \boldsymbol{g}_H, \boldsymbol{g}$) ▷ *Transform into the Hermite gauge using (8) and (9)*
  return $\boldsymbol{M}_H, \boldsymbol{\beta}_H, \boldsymbol{g}_H$
end function